Example 9.64

Find the volume of the spherical cap of height h cut of from a sphere of radius r.

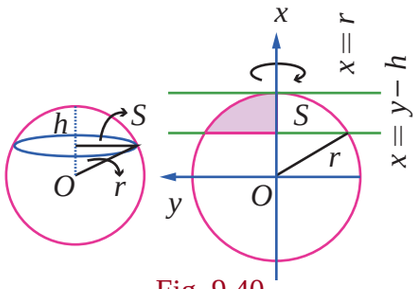


Fig. 9.40

Solution

If the region in the first quadrant bounded by the circle $x^2 + y^2 = r^2$, the x-axis, the lines $x = r - h$ and $x = r$ is revolved about the x-axis, then the solid generated is a spherical cap of height h cut of from a sphere of radius r. See Fig. 9.40. Hence, the required volume is given by

$$V = \pi\int_{r-h}^{r} y^2\,dx = \pi\int_{r-h}^{r}\left(r^2 - x^2\right)dx = \pi\left(r^2x - \frac{x^3}{3}\right)_{r-h}^{r}$$

$$= \pi\left(r^2\left(r-\left(r-h\right)\right) - \frac{\left(r^3 - \left(r-h\right)^3\right)}{3}\right) = \pi\left(r^2h - \frac{\left(r^3 - \left(r^3 - 3r^2h + 3rh^2 - h^3\right)\right)}{3}\right)$$

$$= \pi\left(\frac{3rh^2 - h^3}{3}\right) = \frac{1}{3}\pi h^2\left(3r - h\right).$$

Note

We can rewrite the above volume in terms of the radius of the cap.

If ρ is the radius of the cap, then $\rho^2 + \left(r-h\right)^2 = r^2$.

Then, we have $r = \dfrac{\rho^2 + h^2}{2h}$. Eliminating r, we get

$$V = \frac{1}{3}\pi h^2\left[3\left(\frac{\rho^2 + h^2}{2h}\right) - h\right] = \frac{1}{3}\pi h\left[\left(\frac{3\rho^2 + h^2}{2}\right)\right] = \frac{1}{6}\pi h\left(3\rho^2 + h^2\right).$$

Example 9.65

Find the volume of the solid formed by revolving the region bounded by the parabola $y = x^2$, x-axis, ordinates $x = 0$ and $x = 1$ about the x-axis.

Solution

The region to be revolved about the x-axis is sketched as in Fig.9.41. Hence, the required volume is given by

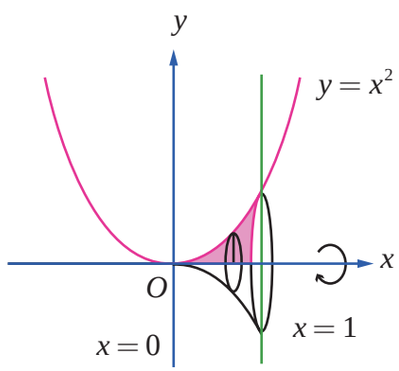


Fig. 9.41

$$V = \pi\int_0^1 y^2\,dx$$

$$= \pi\int_0^1 \left(x^2\right)^2\,dx$$

$$= \pi\left[\frac{x^5}{5}\right]_0^1$$

$$= \pi\left[\frac{1}{5} - 0\right] = \frac{\pi}{5}.$$

Example 9.66

Find the volume of the solid formed by revolving the region bounded by the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1, a > b$ about the major axis.